Linear Transformation -PCA

CSCI-347 Data Mining

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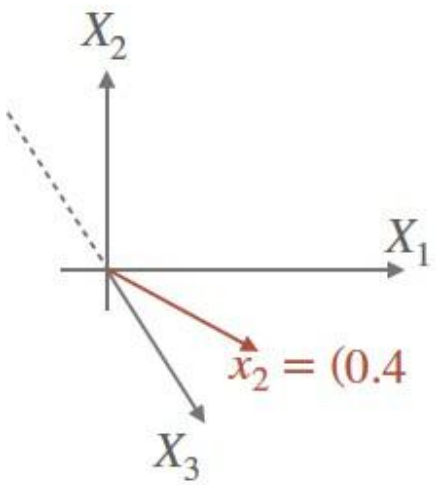
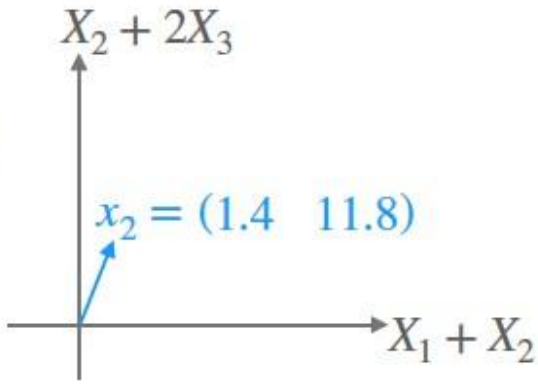
How can we find the best low-dimensional representation of the data?

- Dimensionality reduction can be accomplished by:
  - Selecting a subset of attributes of the dataset.
    - What are the possible advantages and disadvantages?
  - Using all the attributes to create new attributes that can represent the data



$$D = \begin{array}{c|ccc} & X_1 & X_2 & X_3 \\ \hline x_1 & 0.2 & 23 & 5.7 \\ x_2 & 0.4 & 1 & 5.4 \\ x_3 & 1.8 & 0 & 5.2 \\ x_4 & 5.6 & 50 & 5.1 \\ x_5 & -0.5 & 34 & 5.3 \\ x_6 & 0.4 & 19 & 5.4 \\ x_7 & 1.1 & 11 & 5.5 \end{array} \rightarrow \begin{array}{c|cc} & X_1+X_2 & X_2+2X_3 \\ \hline x_1 & 23.2 & 34.4 \\ x_2 & 1.4 & 11.8 \\ x_3 & 1.8 & 10.4 \\ x_4 & 55.6 & 60.2 \\ x_5 & 33.5 & 44.6 \\ x_6 & 19.4 & 29.8 \\ x_7 & 12.1 & 22 \end{array}$$

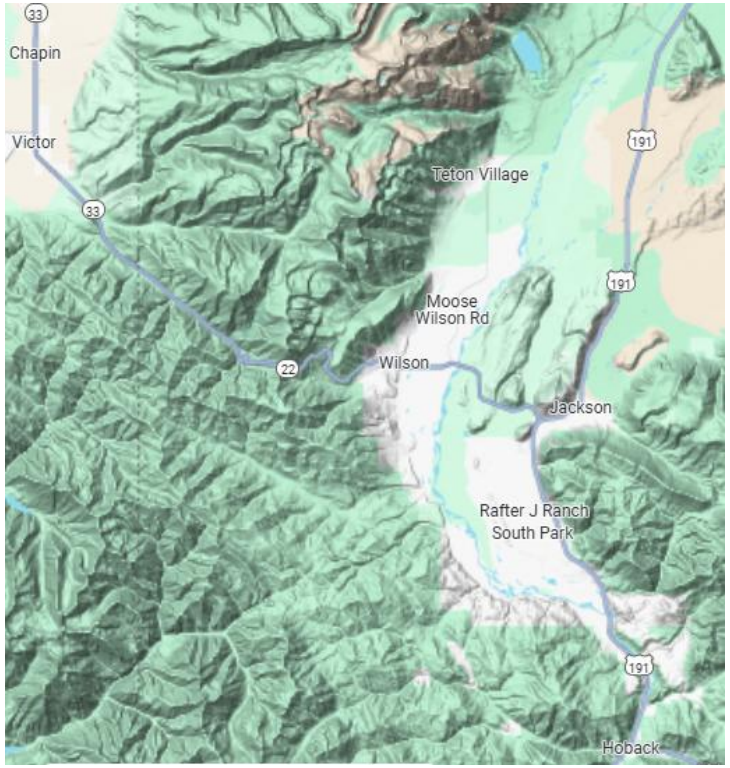


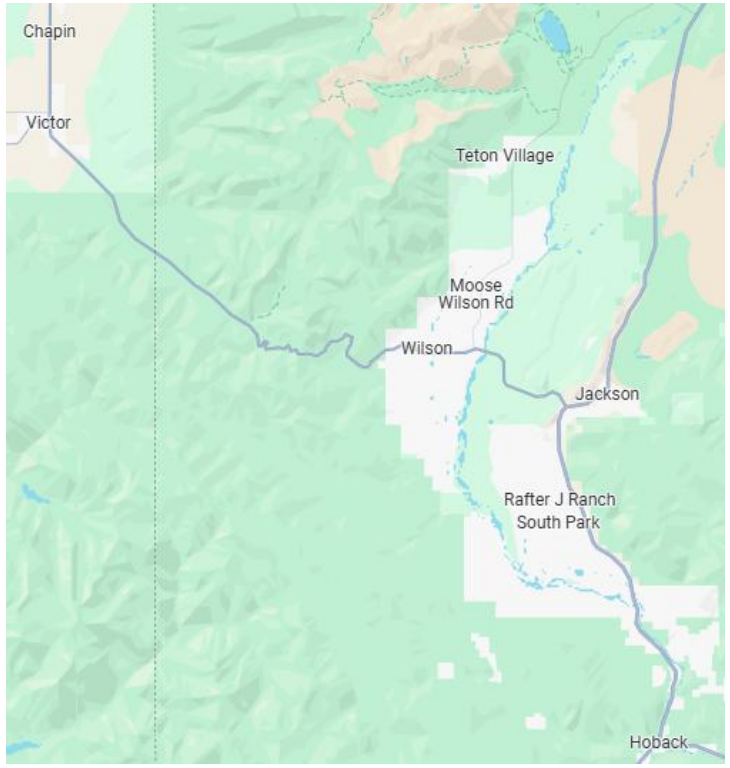
Dimension Reduction

- The goal of dimensionality reduction is to find lower dimensional representation of dataset (Matrix) D to avoid **curse of dimensionality**.
- **Essentially, we are looking for technique to reduce the dimension of a $n \times p$ data matrix to $n \times m$ where $m \leq p$.**
- **Let's look at an example.**

Dimension Reduction

- The idea is to replace large number of variables in the data with a smaller number while maintaining a good representation of the data.





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Principal Component Analysis (PCA)

- Principal Component Analysis (PCA) is the most common dimension reduction technique.

Principal Component Analysis (PCA)

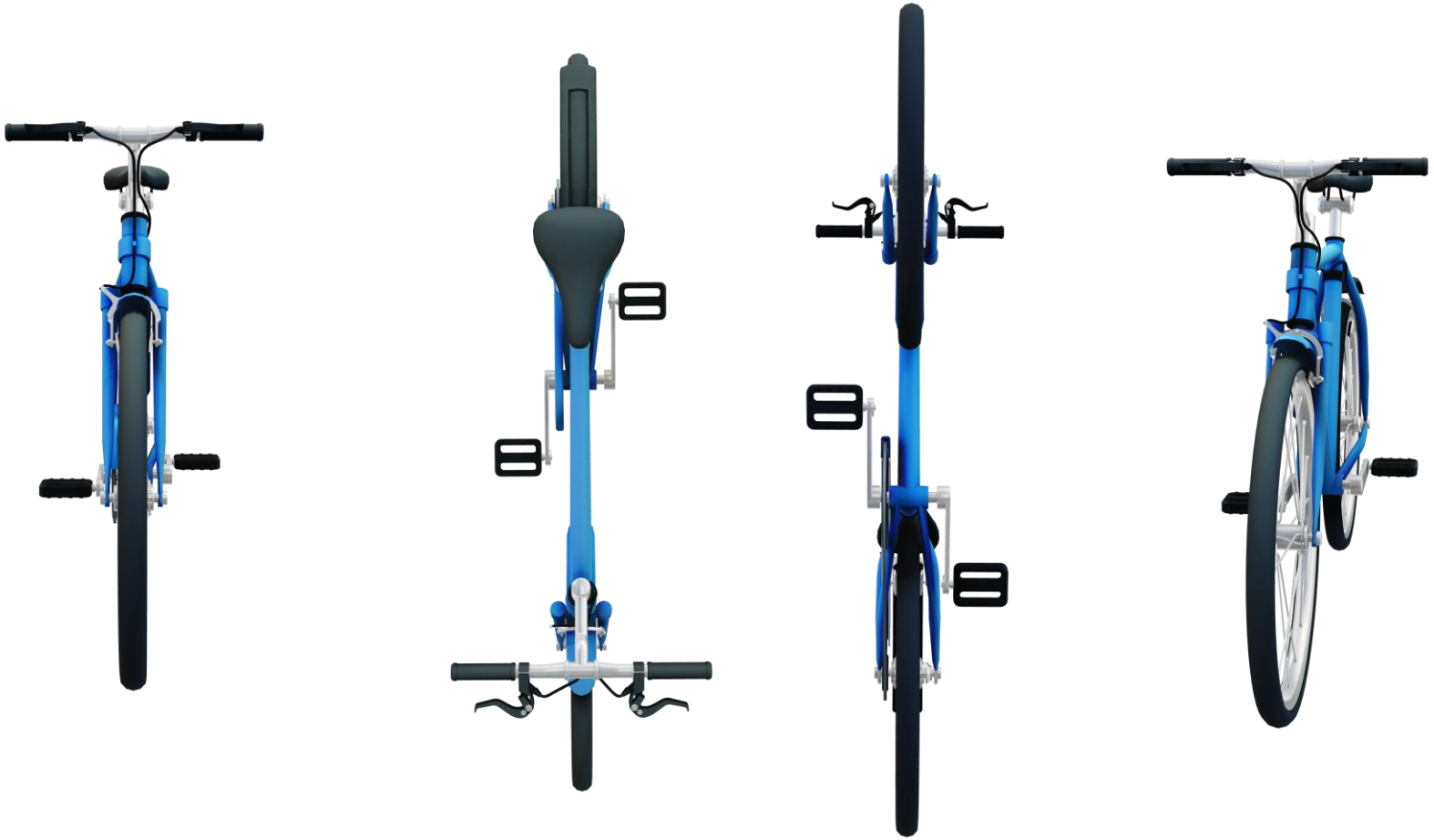
- Let’s try to draw a simple diagram of a bicycle in 2D plane.

Principal Component Analysis (PCA)

- Most of you would draw something like this.

Principal Component Analysis (PCA)

- But you wouldn’t look at the bicycle in the following angles to draw.



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Principal Component Analysis (PCA)

- When you draw a diagram of a bicycle , we are projecting a 3D object onto a 2D plane. Most people would draw the diagram of the bicycle along its most elongated axes. Such projection captures lot more information about the bicycle than other angles.

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Principal Component Analysis (PCA)

- The same idea is used in PCA.
- PCA is mathematical technique that allows you to find a desirable projection that reduces the number of features (attributes) in a matrix while preserving as much information as possible.
- The way PCA achieves this goal is by finding the directions in feature space along which the data has the largest spread or variability.

Principal Component Analysis (PCA)

- PCA produces a low-dimensional representation of a dataset.
- PCA finds a sequence of linear combinations of the variables that have maximal variance and are mutually uncorrelated.
- What can we do with PCA?
- Produce derived variables for use in supervised learning problems (Feature selection)
- Serves as a tool for data visualization.

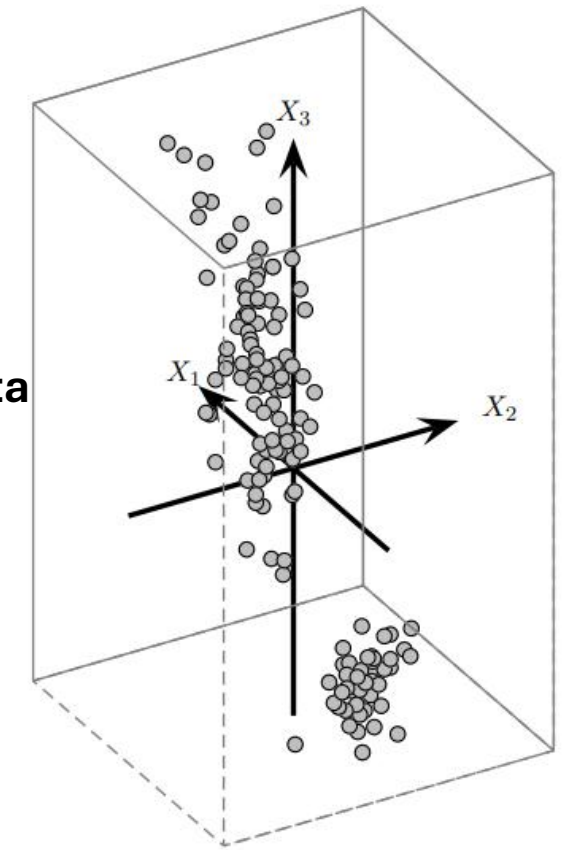
Principal Component Analysis

- The first principal component direction of the data is the direction which the observations vary the most or the direction with the largest variance.
- The second principal component maximize variance in the subspace orthogonal to the first principal component.
- Thus, we basically replace original variables (features/attributes) that may have been correlated with a small set of principal components that capture their joint variation.

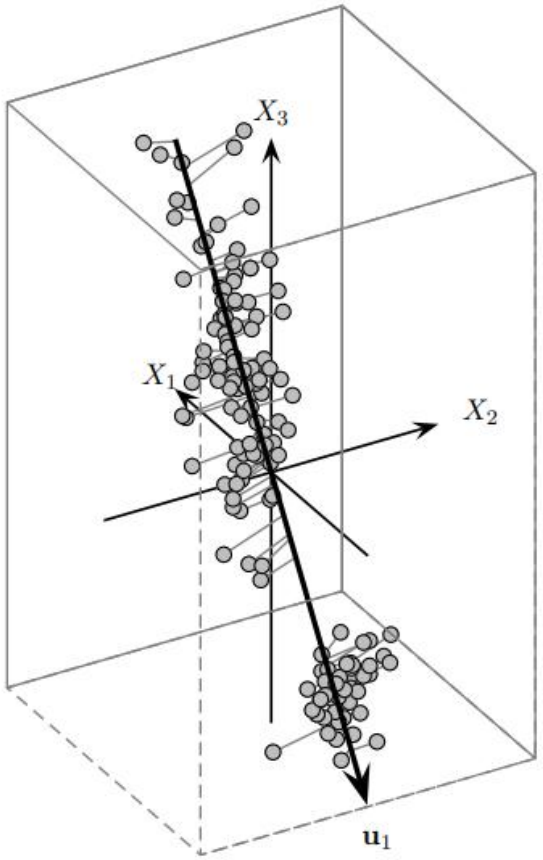
Iris Data: Optimal One-Dimensional basis

- The direction that maximizes the variance is also the one that minimizes the squared error.

Iris Data



Iris Data: Optimal 1D Basis



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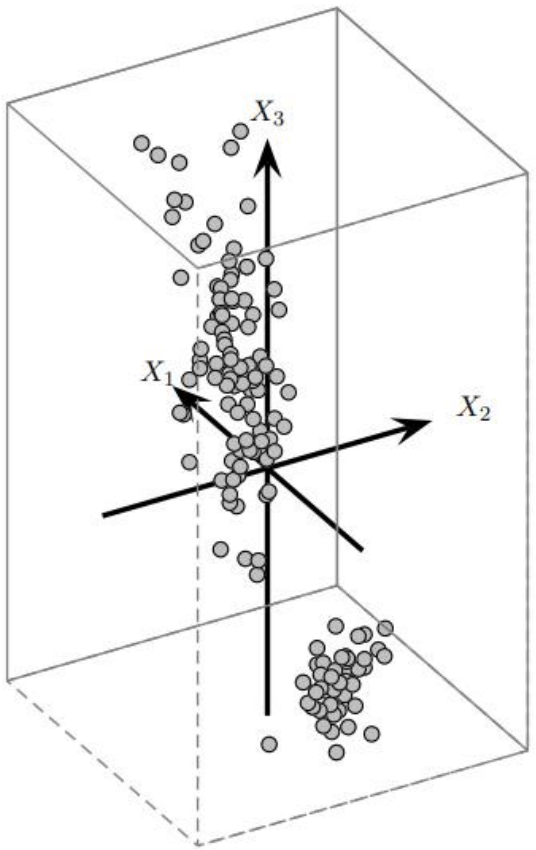
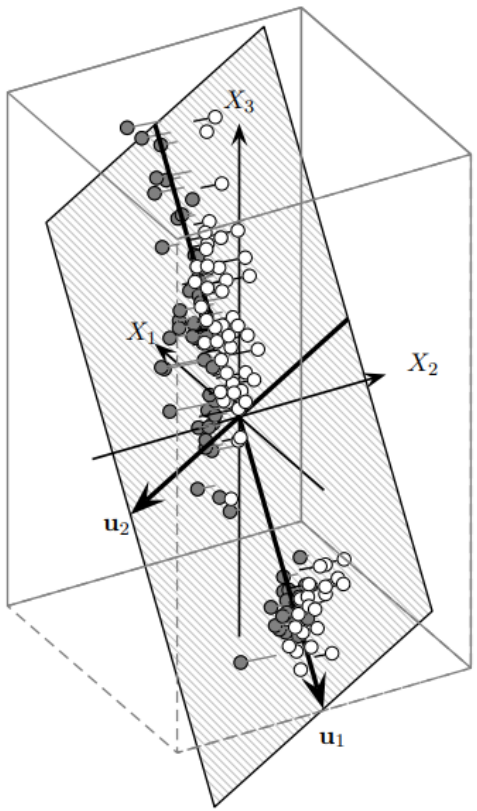
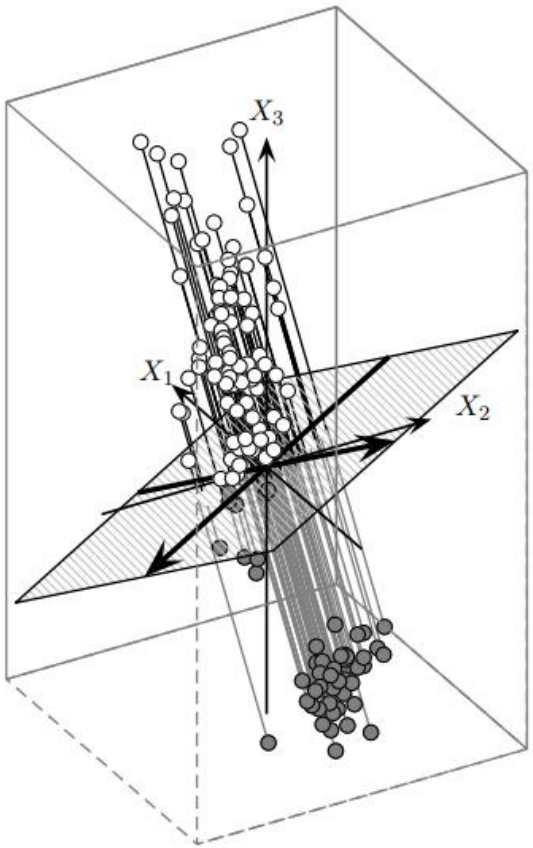




Iris Data Optimal 2D Basis

Iris Data: Non-Optimal 2D Basis



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Summary: PCA

- Principal Component Analysis (PCA) is a technique that seeks a r-dimensional basis that best captures the variance in the dataset.
- The direction with the largest projected variance is called the first principal component.
- The orthogonal direction that captures the second largest projected variance is called the second principal component, and so on.
- The direction that maximizes the variance is also the one that minimizes the mean squared errors.

Linear transformation

- Linear transformation is a function between two vector spaces
  - $T: V \to W$
- Such that the following properties hold for all vectors $u, v \in V$ and scalars c:

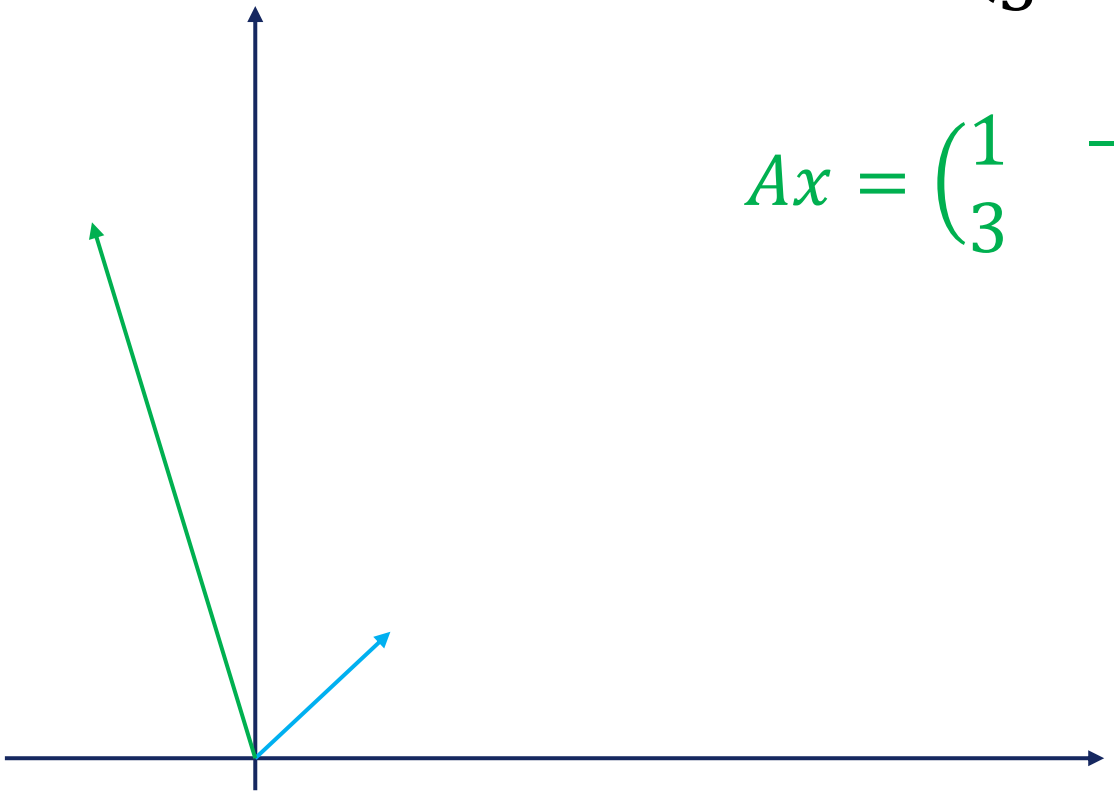
1. $T(u + v) = T(u) + T(v)$
2. $T(cu) = cT(u)$

Linear Transformations

- Different A matrices corresponds to different linear combinations:

$$A = \begin{pmatrix} 1 & -2 \\ 3 & 2 \end{pmatrix}, x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$Ax = \begin{pmatrix} 1 & -2 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 5 \end{pmatrix}$$

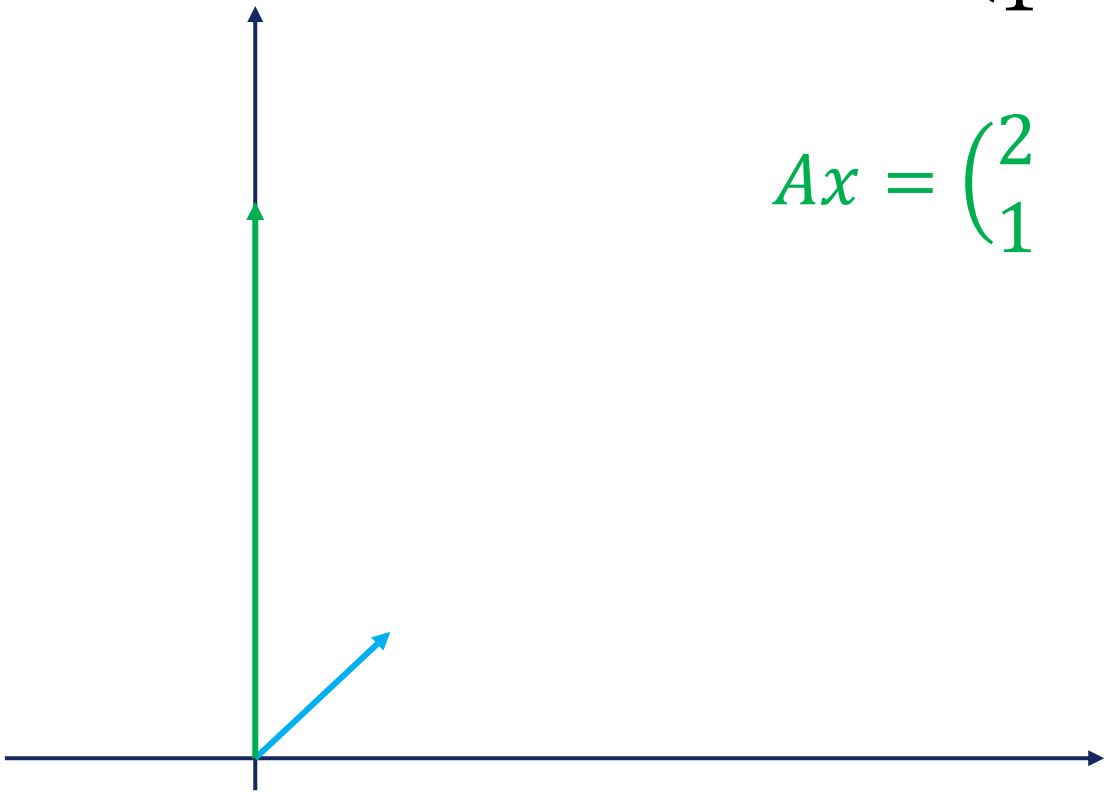


Linear Transformations

- Different A matrices corresponds to different linear combinations:

$$A = \begin{pmatrix} 2 & -2 \\ 1 & 2 \end{pmatrix}, x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$Ax = \begin{pmatrix} 2 & -2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 5 \end{pmatrix}$$



Linear Transformations

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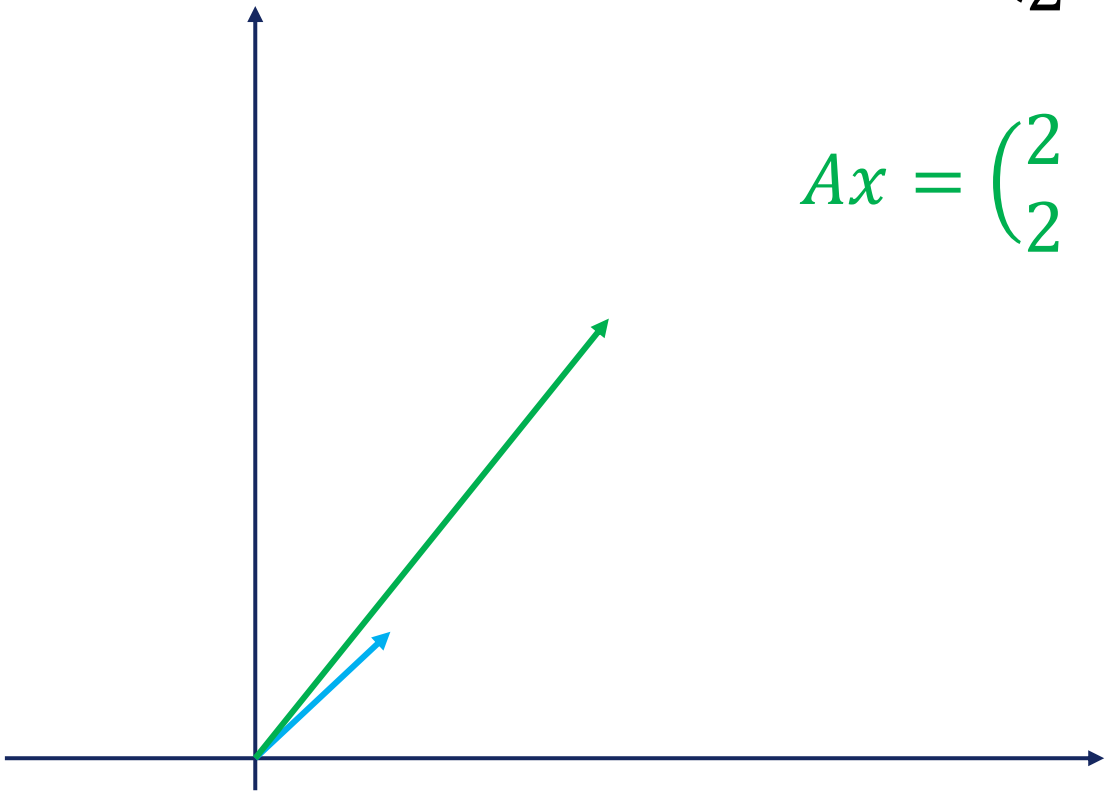
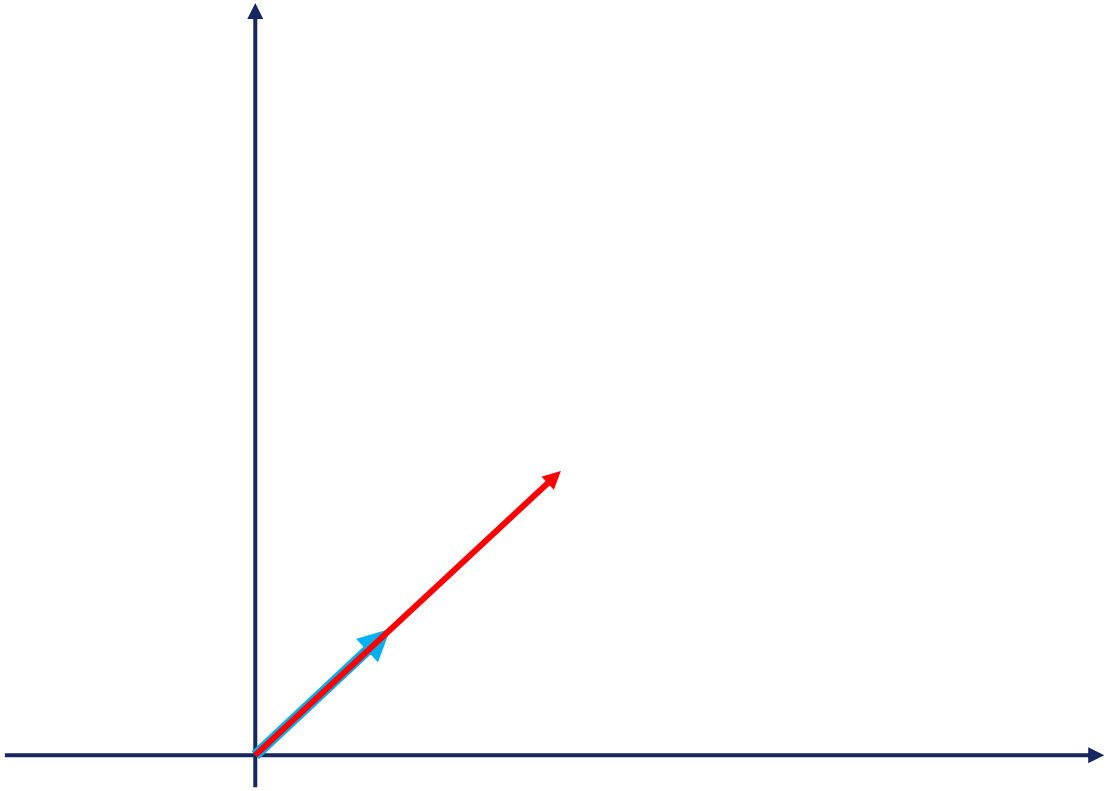
$$Ax = \begin{pmatrix} 2 & 1 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$



Eigenvalues and eigenvectors

- The special vectors that, when multiplied by A, it only change magnitude, are A's eigenvectors and the change in magnitude is the eigenvalue:



$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$Ax = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3 \cdot x$$

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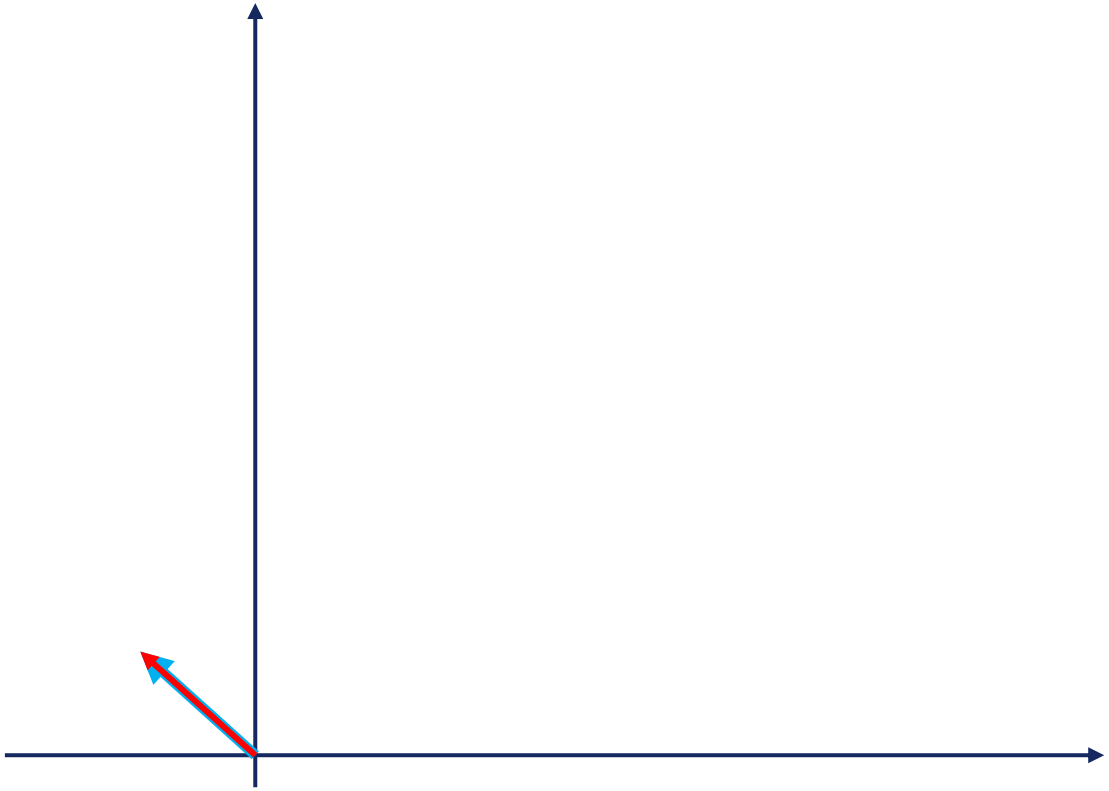

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$Ax = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} = 1 \cdot x$$

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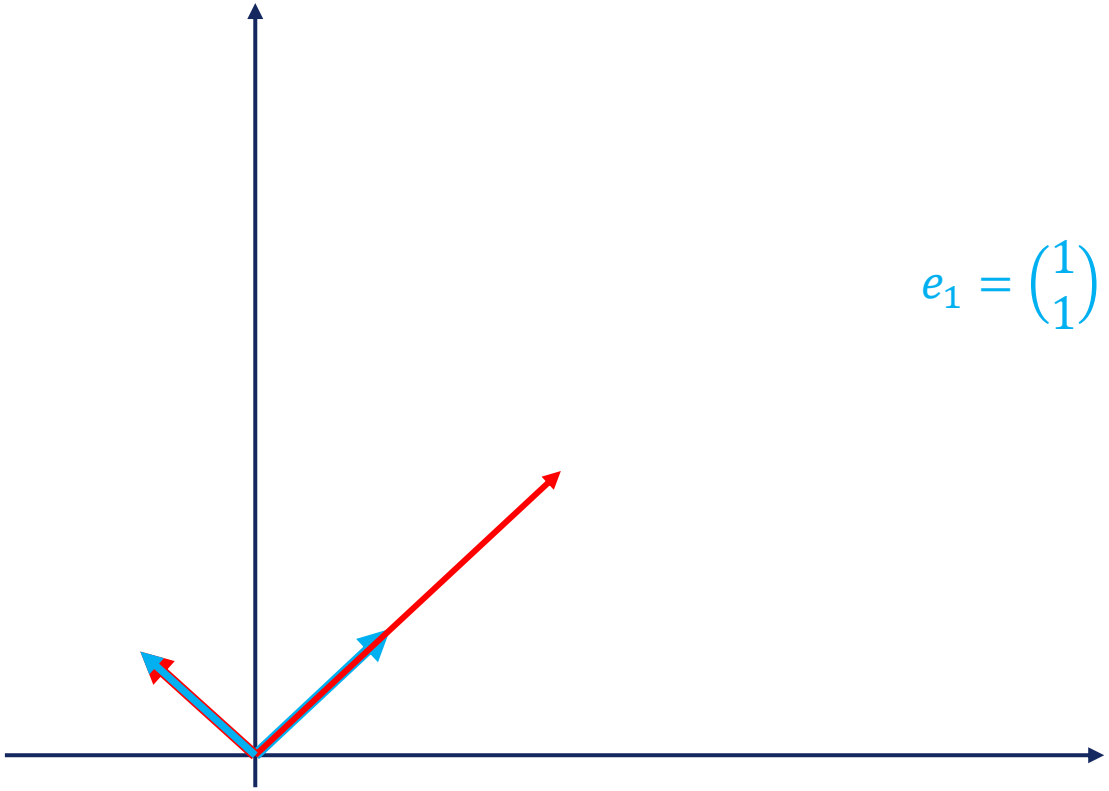

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

$$e_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad e_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \quad Ae_1 = \lambda_1 e_1 \quad Ae_2 = \lambda_2 e_2$$

Linear transformation

- Consider the following data matrix and linear transformation of each data instance.

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \quad x_1 = \begin{pmatrix} 0.2 \\ 23 \end{pmatrix} \quad Ax_1 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 0.2 \\ 23 \end{pmatrix} = \begin{pmatrix} 23.4 \\ 46.4 \end{pmatrix}$$

	X_1	X_2
x_1	0.2	23
x_2	0.4	1
x_3	1.8	0.2
x_4	5.6	50
x_5	-0.5	34
x_6	0.4	19
x_7	1.1	11

$\longrightarrow$

	$2X_1 + X_2$	$2X_1 + 2X_2$
x_1	23.4	46.4
x_2	1.8	2.8
x_3	4.1	4.6
x_4	61.2	111.2
x_5	33	67
x_6	19.8	38.3
x_7	13.2	24.2

PCA, Eigenvector and Eigenvalues

Let's put them together

- The first principal component direction of the data is that along which the observations vary the most. These directions are precisely what we call the eigenvectors of the data's covariance matrix.
- Once we identify these directions (eigenvectors), we need to know how significant or important each direction is. This is where we use eigenvalues to rank each eigenvector. Eigenvalue associated with each eigenvector tells us how much variance is associated along each eigenvector.
  - Larger the eigenvalue, larger variance is captured by the eigenvector.
- In PCA, we rank these eigenvectors (and their corresponding eigenvalues) from most important to least important. By selecting the top few eigenvectors, we can project our original dataset onto simpler, smaller, shadow that still captures the structure of the dataset.

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Transformations using Eigenvectors of Σ

- Start with mean centered dataset.
- Find the covariance matrix Σ.
- Find the eigenvectors and eigenvalues of the covariance matrix Σ.
- Generate the linearly transformed dataset (matrix), by projecting each data instance to the chosen eigenvectors.
- We can also see that the covariance matrix of this transformed dataset shows that there is no covariance between attributes.

PCA Algorithms

Let’s go through an example.

PCA Algorithm

$PCA(D, \alpha)$

1. $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$

$$\Sigma = \begin{pmatrix} 4.14 & 18.42 \\ 18.42 & 321.32 \end{pmatrix}$$

	X_1	X_2
x_1	0.2	23
x_2	0.4	1
x_3	1.8	0.2
x_4	5.6	50
x_5	-0.5	34
x_6	0.4	19
x_7	1.1	11

$$\mu = (1.29 \quad 19.79)$$

PCA Algorithm

$PCA(D, \alpha)$

1. $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$
2. $Z = D - 1_n \cdot \mu^T \quad (1 \in R^{n\times 1})$

	Z_1	Z_2
z_1	-1.1	3.2
z_2	-0.9	-18.8
z_3	0.5	-19.3
z_4	4.3	30.2
z_5	-1.8	14.2
z_6	-0.9	-0.8
z_7	-0.2	-8.8

$\mu_z = (0 \quad 0)$

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z_4	4.3	30.2
z_5	-1.8	14.2
z_6	-0.9	-0.8
z_7	-0.2	-8.8

The total variance is $4.14 + 321.32 = 325.46$

$$\mu = (0 \quad 0)$$

$$\Sigma = \begin{pmatrix} 4.14 & 18.42 \\ 18.42 & 321.32 \end{pmatrix}$$

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3. $\Sigma = \frac{1}{n-1} \cdot Z^T Z$
4. $(\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_d)$
5. $(u_1, u_2, u_3, \dots, u_d)$

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$$\mu = (0 \quad 0)$$

$$\Sigma = \begin{pmatrix} 4.14 & 18.42 \\ 18.42 & 321.32 \end{pmatrix}$$

$$(\lambda_1 \quad \lambda_2) = (322.39 \quad 3.08)$$

$$(u_1 \quad u_2) = \begin{pmatrix} -0.058 & 0.998 \\ -0.998 & -0.058 \end{pmatrix}$$

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4. $(\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_d)$
5. $(u_1, u_2, u_3, \dots, u_d)$
6. Choose the smallest r such that $f(r) \geq \alpha$, where:

$$f(r) = \frac{\sum_{i=1}^{r} \lambda_i}{\sum_{i=1}^{d} \lambda_i}$$

This is the fraction of total variance that is preserved in the first r principal components!

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z_1	-1.1	3.2
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$$f(1) = \frac{\lambda_1}{\sum_{i=1}^{2} \lambda_i} = \frac{322.39}{322.29 + 3.08} = 0.991$$

$$f(2) = \frac{\lambda_1 + \lambda_2}{\sum_{i=1}^{2} \lambda_i} = \frac{322.39 + 3.08}{322.29 + 3.08} = 1$$

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5. $(u_1, u_2, u_3, \dots, u_d)$
6. Choose the smallest r such that $f(r) \geq \alpha$, where:

$$f(r) = \frac{\sum_{i=1}^{r} \lambda_i}{\sum_{i=1}^{d} \lambda_i}$$

This means over 99% of the variance is captured in the direction of the first principal component

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6. Choose the smallest r such that $f(r) \geq \alpha$, where:

$$f(r) = \frac{\sum_{i=1}^{r} \lambda_i}{\sum_{i=1}^{d} \lambda_i}$$

If $\alpha = 0.99$ (or smaller), then $r = 1$

This means over 99% of the variance is captured in the direction of the first principal component

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3. $\Sigma = \frac{1}{n-1} \cdot Z^T Z$
4. $(\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_d)$
5. $(u_1, u_2, u_3, \dots, u_d)$
6. Choose the smallest r such that $f(r) \geq \alpha$, where:

$$f(r) = \frac{\sum_{i=1}^{r} \lambda_i}{\sum_{i=1}^{d} \lambda_i}$$

7. Find the new coordinates a_i by linearly transforming the input dataset.

$$f(1) = \frac{\lambda_1}{\sum_{i=1}^{2} \lambda_i} = \frac{322.39}{322.29 + 3.08} = 0.991$$

	Z_1	Z_2
z_1	-1.1	3.2
z_2	-0.9	-18.8
z_3	0.5	-19.3
z_4	4.3	30.2
z_5	-1.8	14.2
z_6	-0.9	-0.8
z_7	-0.2	-8.8

$$\mu = (0 \quad 0)$$

$$\Sigma = \begin{pmatrix} 4.14 & 18.42 \\ 18.42 & 321.32 \end{pmatrix}$$

$$(\lambda_1 \quad \lambda_2) = (322.39 \quad 3.08)$$

$$(u_1 \quad u_2) = \begin{pmatrix} -0.058 & 0.998 \\ -0.998 & -0.058 \end{pmatrix}$$

PCA Algorithm

$PCA(D, \alpha)$

1. $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$
2. $Z = D - 1_n \cdot \mu^T \quad (1 \in R^{n\times 1})$
3. $\Sigma = \frac{1}{n-1} \cdot Z^T Z$
4. $(\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_d)$
5. $(u_1, u_2, u_3, \dots, u_d)$
6. Choose the smallest r such that $f(r) \geq \alpha$, where:

$$f(r) = \frac{\sum_{i=1}^{r} \lambda_i}{\sum_{i=1}^{d} \lambda_i}$$

7. Find the new coordinates a_i by linearly transforming the input dataset.

$U_r = (u_1, u_2, \dots, u_r), a_i = \mathrm{U_r^T x_i} \ \ \forall i \in [n]$

	Z_1	Z_2
z_1	-1.1	3.2
z_2	-0.9	-18.8
z_3	0.5	-19.3
z_4	4.3	30.2
z_5	-1.8	14.2
z_6	-0.9	-0.8
z_7	-0.2	-8.8

$\mu = (0 \quad 0) \qquad \Sigma = \begin{pmatrix} 4.14 & 18.42 \\ 18.42 & 321.32 \end{pmatrix}$

$(\lambda_1 \quad \lambda_2) = (322.39 \quad 3.08)$

$(u_1 \quad u_2) = \begin{pmatrix} -0.058 & 0.998 \\ -0.998 & -0.058 \end{pmatrix}$

$f(1) = \frac{\lambda_1}{\sum_{i=1}^{2} \lambda_i} = \frac{322.39}{322.29 + 3.08} = 0.991$

$\boldsymbol{U_1 = (u_1) = \begin{pmatrix} -0.058 \\ -0.998 \end{pmatrix}}$

$\boldsymbol{U_1^T = (u_1^T)}$
$\boldsymbol{= (-0.058 \quad -0.998)}$

$D_{transformed}=$

	$\boldsymbol{-0.058X_1 - 0.998X_2}$
$\boldsymbol{a_1}$	$\boldsymbol{\mathrm{U_1^T x_1}}$
$\boldsymbol{a_2}$	$\boldsymbol{U_1^T x_2}$
$\boldsymbol{a_3}$	$\boldsymbol{U_1^T x_3}$
$\boldsymbol{a_4}$	$\boldsymbol{U_1^T x_4}$
$\boldsymbol{a_5}$	$\boldsymbol{U_1^T x_5}$
$\boldsymbol{a_6}$	$\boldsymbol{U_1^T x_6}$
$\boldsymbol{a_7}$	$\boldsymbol{U_1^T x_7}$

PCA Algorithm

$PCA(D, \alpha)$

1. $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$
2. $Z = D - 1_n \cdot \mu^T \quad (1 \in R^{n\times 1})$
3. $\Sigma = \frac{1}{n-1} \cdot Z^T Z$
4. $(\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_d)$
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z_1	-1.1	3.2
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z_3	0.5	-19.3
z_4	4.3	30.2
z_5	-1.8	14.2
z_6	-0.9	-0.8
z_7	-0.2	-8.8

$$\mu = (0 \quad 0) \qquad \Sigma = \begin{pmatrix} 4.14 & 18.42 \\ 18.42 & 321.32 \end{pmatrix}$$

$$(\lambda_1 \quad \lambda_2) = (322.39 \quad 3.08)$$

$$f(1) = \frac{\lambda_1}{\sum_{i=1}^{2} \lambda_i} = \frac{322.39}{322.29 + 3.08} = 0.991$$

$$(u_1 \quad u_2) = \begin{pmatrix} -0.058 & 0.998 \\ -0.998 & -0.058 \end{pmatrix}$$

$$U_1 = (u_1) = \begin{pmatrix} -0.058 \\ -0.998 \end{pmatrix}$$

$$U_1^T = (u_1^T) = (-0.058 \quad -0.998)$$

$D_{transformed}=$

	$-0.058X_1 - 0.998X_2$
a_1	$-0.058(-1.1) - 0.998(3.2)$
a_2	$-0.058(-0.9) - 0.998(-18.8)$
a_3	$-0.058(0.5) - 0.998(-19.3)$
a_4	$-0.058(4.3) - 0.998(30.2)$
a_5	$-0.058(-1.8) - 0.998(14.2)$
a_6	$-0.058(-0.9) - 0.998(-0.8)$
a_7	$-0.058(-0.2) - 0.998(-8.8)$

PCA Algorithm

$PCA(D, \alpha)$

1. $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$
2. $Z = D - 1_n \cdot \mu^T \quad (1 \in R^{n\times 1})$
3. $\Sigma = \frac{1}{n-1} \cdot Z^T Z$
4. $(\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_d)$
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6. Choose the smallest r such that $f(r) \geq \alpha$, where:

$$f(r) = \frac{\sum_{i=1}^{r} \lambda_i}{\sum_{i=1}^{d} \lambda_i}$$

7. Find the new coordinates a_i by linearly transforming the input dataset.

$U_r = (u_1, u_2, \dots, u_r)$, $a_i = \mathrm{U_r^T x_i} \ \ \forall i \in [n]$

$$a_2 = U_1^T x_2 = (-0.058 \quad -0.998)\begin{pmatrix} -0.9 \\ -18.8 \end{pmatrix}$$

$$f(1) = \frac{\lambda_1}{\sum_{i=1}^{2} \lambda_i} = \frac{322.39}{322.29 + 3.08} = 0.991$$

$$U_1 = (u_1) = \begin{pmatrix} -0.058 \\ -0.998 \end{pmatrix}$$

$$U_1^T = (u_1^T) = (-0.058 \quad -0.998)$$

	Z_1	Z_2
z_1	-1.1	3.2
z_2	-0.9	-18.8
z_3	0.5	-19.3
z_4	4.3	30.2
z_5	-1.8	14.2
z_6	-0.9	-0.8
z_7	-0.2	-8.8

$$\mu = (0 \quad 0) \qquad \Sigma = \begin{pmatrix} 4.14 & 18.42 \\ 18.42 & 321.32 \end{pmatrix}$$

$$(\lambda_1 \quad \lambda_2) = (322.39 \quad 3.08)$$

$$(u_1 \quad u_2) = \begin{pmatrix} -0.058 & 0.998 \\ -0.998 & -0.058 \end{pmatrix}$$

$D_{transformed}=$

	$-0.058X_1 - 0.998X_2$
a_1	$-0.058(-1.1) - 0.998(3.2)$
a_2	$-0.058(-0.9) - 0.998(-18.8)$
a_3	$-0.058(0.5) - 0.998(-19.3)$
a_4	$-0.058(4.3) - 0.998(30.2)$
a_5	$-0.058(-1.8) - 0.998(14.2)$
a_6	$-0.058(-0.9) - 0.998(-0.8)$
a_7	$-0.058(-0.2) - 0.998(-8.8)$

PCA Algorithm

$PCA(D, \alpha)$

1. $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$
2. $Z = D - 1_n \cdot \mu^T \quad (1 \in R^{n\times 1})$
3. $\Sigma = \frac{1}{n-1} \cdot Z^T Z$
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	Z_1	Z_2
z_1	-1.1	3.2
z_2	-0.9	-18.8
z_3	0.5	-19.3
z_4	4.3	30.2
z_5	-1.8	14.2
z_6	-0.9	-0.8
z_7	-0.2	-8.8

$$\mu = (0 \quad 0) \qquad \Sigma = \begin{pmatrix} 4.14 & 18.42 \\ 18.42 & 321.32 \end{pmatrix}$$

$$(\lambda_1 \quad \lambda_2) = (322.39 \quad 3.08)$$

$$f(1) = \frac{\lambda_1}{\sum_{i=1}^{2} \lambda_i} = \frac{322.39}{322.29 + 3.08} = 0.991$$

$$(u_1 \quad u_2) = \begin{pmatrix} -0.058 & 0.998 \\ -0.998 & -0.058 \end{pmatrix}$$

$$U_1 = (u_1) = \begin{pmatrix} -0.058 \\ -0.998 \end{pmatrix}$$

$$U_1^T = (u_1^T) = (-0.058 \quad -0.998)$$

$D_{transformed}=$

	$-0.058X_1 - 0.998X_2$
a_1	-22.97
a_2	-1.02
a_3	-0.60
a_4	-50.22
a_5	-33.90
a_6	-18.99
a_7	-11.04

PCA Algorithm

$PCA(D, \alpha)$

1. $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$
2. $Z = D - 1_n \cdot \mu^T \quad (1 \in R^{n\times 1})$
3. $\Sigma = \frac{1}{n-1} \cdot Z^T Z$
4. $(\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_d)$
5. $(u_1, u_2, u_3, \dots, u_d)$
6. Choose the smallest r such that $f(r) \geq \alpha$, where:

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7. Find the new coordinates a_i by linearly transforming the input dataset.

$U_r = (u_1, u_2, \dots, u_r)$, $a_i = \mathrm{U_r^T x_i} \ \forall i \in [n]$

Now we have a 1-dimensional representation of original data that has almost all the variance of the 2-D dataset.

$$f(1) = \frac{\lambda_1}{\sum_{i=1}^{2} \lambda_i} = \frac{322.39}{322.29 + 3.08} = 0.991$$

$$U_1 = (u_1) = \begin{pmatrix} -0.058 \\ -0.998 \end{pmatrix}$$

$$U_1^T = (u_1^T) = (-0.058 \quad -0.998)$$

	Z_1	Z_2
z_1	-1.1	3.2
z_2	-0.9	-18.8
z_3	0.5	-19.3
z_4	4.3	30.2
z_5	-1.8	14.2
z_6	-0.9	-0.8
z_7	-0.2	-8.8

$$\mu = (0 \quad 0) \qquad \Sigma = \begin{pmatrix} 4.14 & 18.42 \\ 18.42 & 321.32 \end{pmatrix}$$

$$(\lambda_1 \quad \lambda_2) = (322.39 \quad 3.08)$$

$$(u_1 \quad u_2) = \begin{pmatrix} -0.058 & 0.998 \\ -0.998 & -0.058 \end{pmatrix}$$

$D_{transformed}$=

	$-0.058X_1 - 0.998X_2$
a_1	-22.97
a_2	-1.02
a_3	-0.60
a_4	-50.22
a_5	-33.90
a_6	-18.99
a_7	-11.04

PCA Algorithm

$PCA(D, \alpha)$

1. $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$
2. $Z = D - 1_n \cdot \mu^T \quad (1 \in R^{n\times 1})$
3. $\Sigma = \frac{1}{n-1} \cdot Z^T Z$
4. $(\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_d)$
5. $(u_1, u_2, u_3, \dots, u_d)$
6. Choose the smallest r such that $f(r) \geq \alpha$, where:

$$f(r) = \frac{\sum_{i=1}^{r} \lambda_i}{\sum_{i=1}^{d} \lambda_i}$$

7. Find the new coordinates a_i by linearly transforming the input dataset.

$U_r = (u_1, u_2, \dots, u_r)$, $a_i = \mathrm{U_r^T x_i} \ \forall i \in [n]$

$$\Sigma = \begin{pmatrix} 4.14 & 18.42 \\ 18.42 & 321.32 \end{pmatrix}$$

Original total variance was 325.46

	Z_1	Z_2
z_1	-1.1	3.2
z_2	-0.9	-18.8
z_3	0.5	-19.3
z_4	4.3	30.2
z_5	-1.8	14.2
z_6	-0.9	-0.8
z_7	-0.2	-8.8

$$\mu = (0 \quad 0)$$

$$f(1) = \frac{\lambda_1}{\sum_{i=1}^{2} \lambda_i} = \frac{322.39}{322.29 + 3.08} = 0.991$$

$$(\lambda_1 \quad \lambda_2) = (322.39 \quad 3.08)$$

$$(u_1 \quad u_2) = \begin{pmatrix} -0.058 & 0.998 \\ -0.998 & -0.058 \end{pmatrix}$$

$$U_1 = (u_1) = \begin{pmatrix} -0.058 \\ -0.998 \end{pmatrix}$$

$$U_1^T = (u_1^T) = (-0.058 \quad -0.998)$$

$var(D_{transformed}) = 322.16$

$D_{transformed}=$

	$-0.058X_1 - 0.998X_2$
a_1	-22.97
a_2	-1.02
a_3	-0.60
a_4	-50.22
a_5	-33.90
a_6	-18.99
a_7	-11.04